

Komodo Dragon Element K

At the start of the semester, we were given the task of creating a Komodo Dragon box for the zoo. First, we had to do some research. We looked at different Komodo Dragon cages and alligator boxes, etc. We also went down to the zoo in order to view their current solution and learn more about what they wanted from the boxes. Once we had created a CAD of our initial prototype, we decided to make a cardboard prototype. This prototype helped us envision how we were going to create the frame of the box.



Once that was done, we created a design matrix that would help us decide what materials to use and what handles to create. Once we had determined the best plan of action, we decided to put in our order. This was when we learned that our budget was cut in half. Not wanting to completely drop the idea, one of our classmates was able to weld some metal for us to use as the frames.



We took time to spray paint, create handles and set up the boxes for when we would deliver them to the zoo. We marked out where holes needed to be drilled so we could attach the plastics, handles, and the connection system. Once that was all done, we had completed the frame of the box.



After we have mainly finished the box, a few more things needed to be done before the boxes were ready. One of the other groups had to work on inserting the medical treatment doors on the plastic sides so that the vets at the Nashville Zoo could use ultrasounds on the Komodo Dragon. The other group had to finish setting up the connection system with butterfly clamps so the boxes can stay together and the Komodo dragon cannot escape from between them.

On May 1, 2025, we were given feedback from two Herpetology Keepers at the Nashville Zoo, a professor from Vanderbilt, and the CCTE at Williamson County Schools. They helped us and gave us some finishing thoughts that would help us create the final boxes.

On Thursday May 8, we drove the boxes down to the zoo to present them to the Zoo Keepers.